Skill Without Physics: A Mechanistic Audit of Neural Gravity-Wave Parameterizations

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Draft v3.1 — every number traceable to a stamped results file in the released repository

Abstract

Neural-network parameterizations of atmospheric gravity-wave (GW) momentum fluxes increasingly inform climate-model development, and horizontal nonlocality is credited for much of their skill. We present the first mechanistic interpretability study of a parameterization emulator, auditing the released single-column / 3×3-stencil / Attention-U-Net hierarchy of Gupta et al. (2025) with a pre-registered hypothesis funnel: 30 hypotheses generated, 24 screened (15 killed), 1 measured, 5 deferred with reasons, and 1 further killed at held-out confirmation. Three findings emerge. (1) The U-Net's variance and extreme-tail calibration rides substantially on a *position-indexed prior*: convolutional padding lets the network infer absolute position, and rolling the input map—which changes no physics—collapses its flux-variance calibration from 1.10 to 0.33. Roll ensembles attribute up to 62% of variance calibration and 37% of extreme-tail (P99.9) calibration to position (an upper bound), while bulk-quantile calibration is flow-driven; the calibration split replicates on a second held-out month. Notably, pooled-histogram Hellinger distance—the literature's headline distributional metric—is insensitive to this collapse and can even be *improved* by physically absurd variance inflation; quantile-ratio ladders discriminate. (2) The U-Net's causally used context is long-range (median 4 grid cells, $\sim$1200 km at T42) but isotropic and unaligned with ray-traced propagation directions. (3) Under single-aspect grafts of real columns—a protocol whose amplitude family is validated without conditions on a 1D testbed emulator that demonstrably learned its physics (response tracking 0.94 there vs. 0.49 here)—the column models show no filtering response at typical columns, drag that ignores wind rotation, and non-monotone amplitude response. These surrogates achieve much of their offline skill as regime pattern-matchers with geographic priors rather than by encoding wave physics, with direct implications for out-of-distribution trust and for how distributional skill should be measured.

1 Introduction

Atmospheric gravity waves transport momentum from their sources—mountains, convection, jets and fronts—into the middle atmosphere, where their breaking drives circulations that climate models cannot resolve and must parameterize (Fritts & Alexander, 2003). Hand-built GW parameterizations embody strong simplifying assumptions (single-column propagation, steady launch spectra), and a growing literature replaces them with neural networks trained on high-resolution data (Espinosa et al., 2022; Hardiman et al., 2023; Gupta et al., 2024a, 2025a). A headline result of this line is that *horizontal nonlocality*—giving the network a view of neighboring columns or the whole domain—substantially improves offline skill, with the natural and widely voiced hope that networks so equipped learn genuinely nonlocal wave physics such as lateral propagation.

Whether that hope is warranted is a question about *mechanism*. The tools of mechanistic interpretability—probing with controls, activation patching, controlled input interventions, representation analysis—are built for such questions, yet have barely touched scientific surrogates: prior interpretability of GW emulators stops at kernel Fourier analysis (Pahlavan et al., 2024), gradient receptive fields (Pahlavan et al., 2025), and SHAP attributions (Haslauer et al., 2026). To our knowledge (verified by a systematic scan, August 2026), no activation-level or interventional study existed for any subgrid parameterization emulator.

We conduct such a study on the *actual released artifacts* of Gupta et al. (2025a): the hierarchy $\mathrm{M1}$ (single-column MLP), $\mathrm{M2}$ (3×3 stencil), $\mathrm{M3}$ (Attention U-Net), the released checkpoints, and held-out months of the models' test year. Because interpretability is vulnerable to cherry-picking, the entire investigation is organized as a pre-registered hypothesis funnel (Sec. 4): falsifiable statements and kill criteria fixed before each test, adversarial critic passes at stage gates, held-out-month confirmation for surviving claims, and full reporting of all outcomes (Appendix A). Fifteen of 24 screened hypotheses died—including our own pre-registered flagship ("the U-Net's influence pattern follows wave-propagation geometry"; it does not)—and the surviving findings were in several cases discovered *by* the kills.

Contributions.

1. **A position-indexed variance prior** (Sec. 5.2). $\mathrm{M3}$ infers absolute position through zero-padding and uses it as a de facto geography channel: its distinguishing skill—variance/extreme-tail calibration—collapses under longitude rolling (variance ratio $1.10 \to 0.33$), with up to 62% of variance calibration attributable to position, localized causally to the bottleneck ($\sim$48%) and decoder ($\sim$39%), concentrated over climatological hotspots, replicated on a second month, and visible as a $+5\%$ error seam at the map boundary.

2. **The character of the learned nonlocality** (Sec. 5.3). Causal occlusion shows context is used to a median radius of 4 cells ($\sim$1200 km), yet influence maps are isotropic (median axis ratio 1.07) and unaligned with ray-traced propagation; the 3×3 gain is carried by the full multivariate neighborhood pattern—not by gradients, single directions, linear readouts, or any low-rank channel we could identify.

3. **A positive-controlled physics-trust audit** (Sec. 5.4). Under single-aspect grafts of real columns the models show no critical-level filtering at typical columns, wind-rotation-blind drag, and non-monotone amplitude scaling—while a testbed emulator that provably learned its (1D) physics passes the same battery.

4. **Methods findings** (Secs. 5.5, 3): pooled Hellinger distance is insensitive to—and gameable against—exactly the failure it is used to detect; and a reproducibility hazard in the public data stack (normalization-incompatible public data + checkpoints; R^2 of -4 to -81 without our released conversion).

This is an offline study of one released family at T42; Sec. 6 details scope limits, including why some nulls may be resolution-conditional.

2 Related Work

ML parameterization of GWs. Scheme emulation established feasibility and out-of-sample generalization (Espinosa et al., 2022; Hardiman et al., 2023); reanalysis-driven flux prediction introduced the nonlocality hierarchy audited here (Gupta et al., 2024a, 2025a), trained on Helmholtz-decomposed ERA5 fluxes (Gupta et al., 2024b) and evaluated with Hellinger distances; foundation-model fine-tuning followed (Gupta et al., 2025b). Nonlocal inputs for subgrid closures more broadly: Wang et al. (2022).

Interpretability of scientific ML. GW-specific prior art is weight- or attribution-level (Pahlavan et al., 2024, 2025; Haslauer et al., 2026). Mechanistic tools recently reached weather foundation models—SAEs and steering on GraphCast (MacMillan & Ouellette, 2025), probes on Aurora (Richards & Balan, 2025), SAEs on Walrus (Rosenfeld & Sonnewald, 2026), cross-model CKA (Craig et al., 2026)—but no probing, patching, or interventional study of a parameterization emulator preceded this work.

CNN position encoding. Padding leaks absolute position into CNNs (Islam et al., 2020; Kayhan & van Gemert, 2020). Its consequences for scientific surrogates—a memorized geographic prior doing work attributed to physics—were unmeasured; we quantify them causally.

3 Setting: Models, Data, and a Reproducibility Hazard

3.1 The released hierarchy

Inputs are ERA5-derived columns on a T42 Gaussian grid (64×128), 122 model levels (surface to $\sim$45 km): u, v, θ (optionally w), plus scalars $[\mathrm{lat}, \mathrm{lon}, z_s]$ for $\mathrm{M1}/\mathrm{M2}$; outputs are both GW momentum-flux components at all levels (244 channels). Outputs are cube-root transformed and globally standardized,

$$\tilde{F} \;=\; \left((F-\mu_F)/\sigma_F\right)^{1/3}, \qquad \tilde{u} \;=\; (u-\mu_u)/(3\sigma_u), \tag{1}$$

with fixed global constants. $\mathrm{M1}$ is a 6-hidden-layer MLP ($\mathrm{hdim} = 4 \times \mathrm{idim}$, LeakyReLU) on single columns; $\mathrm{M2}$ prepends one 3×3 valid convolution; $\mathrm{M3}$ is an attention-gated U-Net (additive skip *gating*, not self-attention; encoder $64 \to 1024$ channels over four poolings) on the global map, and—consequentially—*omits* the lat/lon/z_s scalars from its inputs. Training used 2010/2012/2014 (all months); all of 2015 is test (from the released training script). We use July 2015 (screening), January 2015 (confirmation; analyses restricted to a four-entry ledger pre-registered before the file was opened), and two released August snapshots. One checkpoint per configuration exists; no seed ensembles were released. Two code-level facts matter later: the checkpoints carry vestigial, never-executed BatchNorm parameters (all `num_batches_tracked` $= 0$), and $\mathrm{M3}$'s convolutions zero-pad a longitudinally *periodic* domain.

3.2 A reproducibility hazard

The anchor's training-format monthly files are published in WxC-Bench. All of them share one *alternative* normalization—different constants and a $1 \times \sigma$ (not $3 \times \sigma$) u, v convention—despite metadata identical in wording to the author-released test files. Fed to the released checkpoints as-is they yield R^2 of -57 to -81 (M1) and -4.1 to -4.4 (M3). We derived the exact affine conversion and validated it three ways: algebraic round-trip; a *consecutive-hour test* (the July file's last hour is adjacent to the released

Table 1: Full-month replication on held-out data (744 hourly timesteps per month). RMSE in normalized space; Hellinger H and variance ratio $\sigma^2_{\hat{F}}/\sigma^2_F$ in physical space. Source: `results/a4_fullmonth`, `a4_january`.

Month	Model	RMSE	$\text{RMSE}_{\cos\varphi}$	H_{uw}	H_{vw}	σ^2-ratio
July 2015	M1	0.7844	0.7882	0.0644	0.1264	0.360
	M2	0.6980	0.6958	0.0493	0.0970	0.363
	M3	0.6775	0.6764	0.0399	0.0433	1.066
paired M2−M3 RMSE: mean 0.0204, 95% CI [0.0202, 0.0207]; M3 better in 100% of 744 timesteps						
January 2015	M1	0.7444	0.7705	0.0699	0.1080	0.419
	M2	0.6653	0.6809	0.0502	0.0914	0.351
	M3	0.6500	0.6661	0.0414	0.0461	0.996
paired M2−M3 RMSE: mean 0.0152, 95% CI [0.0150, 0.0155]; M3 better in 100% of 744 timesteps						

August snapshot's first hour; converted fields match with regression slopes ≈1); and a skill gate (converted data reproduces paper-consistent R^2, e.g. M3 0.37–0.42 in both months). All experiments auto-detect and convert; utilities are released.

3.3 Replication gate

Table 1 replicates the anchor's core offline result on both held-out months: RMSE and Hellinger orderings M3<M2<M1 (robust to $\cos\varphi$ area weighting), with M3 beating M2 in *744/744* paired timesteps in each month. The skill benefit of nonlocality is unambiguous; this paper asks what implements it.

4 A Pre-Registered Audit

Funnel. Thirty falsifiable hypotheses across seven families (information content, representations, architecture, gates, transfer, failure structure, physics consistency), each with technique family, cheapest test, and kill criterion; an adversarial critic pass hardened thresholds and controls before screening. Screening tests were budgeted <20 minutes of CPU; survivors advanced to deep dives under pre-registered analysis plans; January confirmations were limited to the pre-registered ledger. Deviations were logged when they occurred (all appear in the released log). **Multiple comparisons:** every test is reported (Appendix A); verdicts are pre-registered thresholds, not p-values; the funnel table is the full family of outcomes.

Distributional metrics. For flux component F with prediction $\hat{F}$ (physical units), the *quantile-ratio ladder* is

$$R_q \;=\; \frac{Q_q(|\hat{F}|)}{Q_q(|F|)}, \qquad q \in \{0.9,\, 0.99,\, 0.999\}, \tag{2}$$

computed per timestep and summarized by mean with a 2000-resample bootstrap CI over timesteps. Pooled Hellinger uses $H = 1 - \sum_i \sqrt{p_i q_i}$ on common histograms. Variance ratio is $\mathrm{Var}(\hat{F})/\mathrm{Var}(F)$ pooled over the month.

Interventions. (i) *Longitude roll*: T_r shifts the input map r cells (physics-neutral: the domain is periodic; M1 is roll-invariant, M2 roll-equivariant—both are controls). Predictions are unrolled before scoring; the *position share* of a calibration quantity V is

$$S_V \;=\; \left(V_0 - \tfrac{1}{15}\textstyle\sum_r V_r\right) / \, V_0, \quad r \in \{8, 16, \ldots, 120\}, \tag{3}$$

an *upper bound* on position dependence (rolling also moves inputs off-manifold). (ii) *Cumulative roll-patching*: encoder activations from the unrolled forward are roll-aligned and patched into the rolled forward at sites $\mathrm{conv}1..k$; an identity control (patching a run with its own activations) reproduces the base prediction to machine precision. (iii) *Occlusion*: inputs outside radius r of a target column are replaced by monthly climatology; the saturation radius is the smallest r with target-column MSE within 5% of the full-input value. (iv) *Neighborhood clamping/resampling* for M2.

Physics grafts. Single-aspect modifications of real columns under an admissibility rule ($\max_c |(x_c - \mu_c)/\sigma_c| \le 4$ against monthly column statistics): *reflection* creates a directional wind reversal above level z_c,

$$u'(z) = u(z) + \beta\left(2u(z_c) - 2u(z)\right), \quad z \text{ above } z_c, \tag{4}$$

with per-column maximal admissible dose $\beta \in [0.25, 1]$; *amplitude* scales deviations from climatology by $a \in [0.25, 1.5]$; *rotation* rotates (u, v) jointly by ϕ. Responses: suppression $s = 1 - \langle|\hat{F}'|\rangle_{\text{above}}/\langle|\hat{F}|\rangle_{\text{above}}$; Spearman ρ of response vs. a; circular alignment of drag-vector rotation with ϕ.

Positive control. Nulls are meaningful only if the tests detect physics where it exists. We ran the same graft families on the released `qbo1d` testbed emulator (10-layer MLP), whose stochastic 20-wave source physics is exactly computable, giving ground truth $\mathbb{E}[S|u]$ by Monte Carlo (200 draws). The emulator is faithful (median corr. 0.983 to $\mathbb{E}[S|u]$); the *amplitude family validates unconditionally* (median Spearman 0.94 between emulator and true response curves) and is the load-bearing null; the reflection family validates where the true effect is detectable (median corr. 0.75 at relative effect ≥10%; agreement rises with effect size, $r = 0.93$)—a conditioning that cannot be applied in ERA5, where ground truth is unknown (Sec. 6).

Probing. Ridge probes with mandatory controls: random-init network of identical architecture, raw-input baseline, shuffled labels, and spatial train/test splits (disjoint longitude sectors); we report *selectivity* = metric(real) − max(metric(rand-init), metric(input)).

5 Results

5.1 Where the distributional skill lives

Table 2 and Fig. 1 give the calibration ladders. All models compress typical amplitudes severely (July P90 ratios 0.07/0.11/0.28 for M1/M2/M3); calibration improves

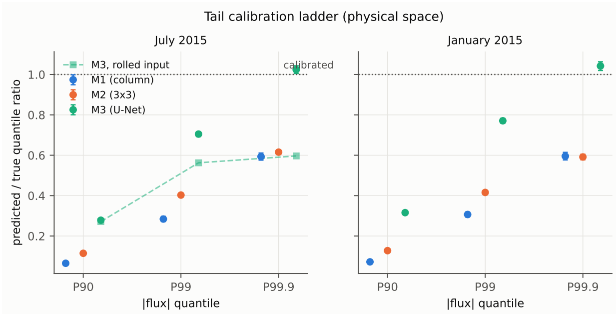

Figure 1: Quantile-ratio calibration ladders (Eq. 2; physical space; bootstrap 95% CIs over timesteps). Left: July, including M3 under a 64-cell input roll—its extreme-tail calibration collapses to the column-model band. Right: January replication. Sources: `screen_a5_tails`, `j1_tails_january`, `gateweek_g2`.

Table 2: Calibration ladders R_q with bootstrap 95% CIs.

	P90	P99	P99.9
July 2015			
M1	0.065 [0.064, 0.066]	0.284 [0.272, 0.297]	0.593 [0.576, 0.611]
M2	0.114 [0.112, 0.116]	0.403 [0.394, 0.412]	0.615 [0.605, 0.626]
M3	0.278 [0.274, 0.283]	0.705 [0.693, 0.716]	1.024 [1.005, 1.043]
M3 (rolled 64)	0.272 [0.263, 0.279]	0.563 [0.533, 0.586]	0.597 [0.547, 0.638]
January 2015			
M1	0.072 [0.070, 0.074]	0.307 [0.294, 0.319]	0.595 [0.577, 0.614]
M2	0.128 [0.125, 0.130]	0.416 [0.408, 0.423]	0.591 [0.577, 0.605]
M3	0.316 [0.310, 0.321]	0.771 [0.758, 0.783]	1.042 [1.020, 1.063]

with quantile level and with nonlocality until M3's extreme tail is calibrated (P99.9 $=$ 1.02, CI $[1.005, 1.043]$). January replicates the pattern in full (M3: 0.32/0.77/1.04). Month-scale variance ratios split the hierarchy sharply: $0.36/0.36/1.07$ (July), $0.42/0.35/1.00$ (January). *The U-Net's distinguishing skill is variance/tail calibration.*

5.2 Finding 1: a position-indexed variance prior

M3 receives no coordinates or orography, yet three causal probes show it knows where it is—and that its calibration advantage depends on it.

Roll test and decomposition. Under T_{64}, M3 degrades globally (+15% RMSE) and its variance calibration collapses to 0.33—*below* M2—with P99.9 falling $1.01 \to 0.60$, while pooled Hellinger is unchanged ($0.0745 \to 0.0744$). The 15-roll ensemble (Fig. 2, Table 3) yields a symmetric distance curve (0.54 at r=8, minimum $\sim$0.33 at half-domain, recovering to 0.56 near full circle—an internal falsification check that passes) and position shares (Eq. 3) of 62% (variance), 37% (P99.9), 17% (P99), 2.5% (P90): *the position prior supplies extreme-tail variance specifically; bulk calibration is flow-driven.* Regionally, the base variance ratio is 1.29 over the eight climatological hotspot boxes vs. 1.02 elsewhere, collapsing to $0.24/0.37$ under roll: the prior lives where the hotspots are.

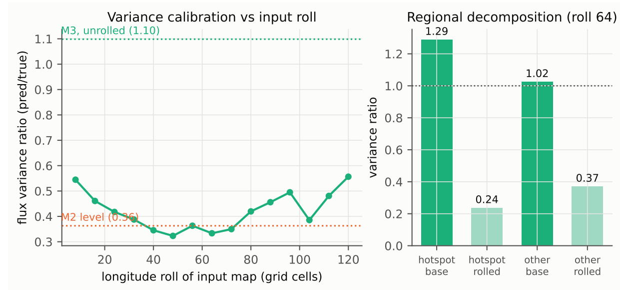

Figure 2: Left: M3 variance ratio vs. input roll distance (12 eval timesteps per roll); symmetry about the half-domain and recovery near the full period tie the collapse to *position*, not to generic input perturbation. Right: regional decomposition at roll 64. Source: `d1_rollensemble`.

Table 3: Roll-ensemble decomposition and causal localization.

Quantity	Unrolled	Roll ensemble (15 rolls)	Position share (upper bd.)
σ^2-ratio	1.098	0.421 ± 0.073	62%
P90	0.280	0.273 ± 0.004	2%
P99	0.697	0.576 ± 0.017	17%
P99.9	1.010	0.641 ± 0.047	37%

Cumulative roll-patch restoration of σ^2-ratio: conv1–3 $\leq$1%, conv4 12%, conv5 61%, decoder (residual) $\approx$39%

Causal localization. Cumulative roll-patching restores essentially none of the calibration through encoder stages 1–3 ($-0.0/-0.0/-0.9$%), $+12$% at conv4 and $+61$% cumulative at the bottleneck (Fig. 3); the residual $\sim$39% returns only when the decoder also sees consistent features. The identity control is exact (max deviation 0.0), so this is not implementation error: *the position computation forms where receptive fields become global and is applied again through decoder-stage boundary effects*, consistent with depth-accumulating padding information (Islam et al., 2020). (Our pre-registered sanity endpoint wrongly assumed full encoder patching restores $\approx$100%; the measured 61% is a finding—the decoder itself computes position-sensitively—and the corrected assumption is logged.)

The seam. A visible corollary (Fig. 4): M3 alone carries $+5.1$% (July) / $+5.05$% (January) excess error at the three columns adjacent to the map boundary (M1: -1.3%, M2: -0.8%); under rolling the absolute excess moves with the padding boundary.

Not reconstruction; not rescaling. Probes for z_s from M3's first encoder stage read *worse* than raw inputs (R^2 0.06 vs. 0.45; random-init 0.08): the encoder *discards* linearly readable orography rather than reconstructing it, and what little z_s information exists at that depth is flow-derived, not position-derived (roll-arm probe: R^2 0.14 vs. source geography, -0.03 vs. map position). Nor is the calibration a trivial output rescaling: per-channel variance-matching gains on M2 explode physical tails (P99 ratio 3.19; variance 11.9; RMSE 1.14) instead of reproducing M3's ladder.

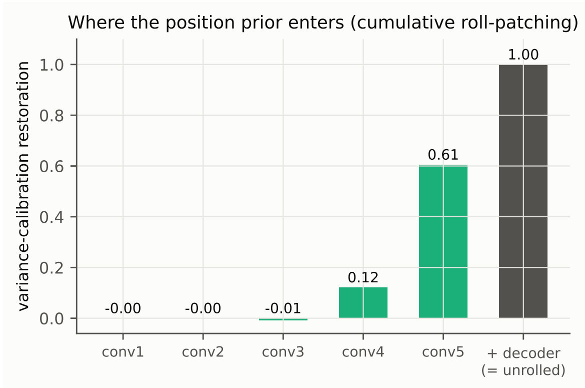

Figure 3: Cumulative roll-patching: fraction of variance calibration restored when encoder activations up to a given stage are replaced by roll-aligned unrolled activations. Source: `d1_rollpatch`.

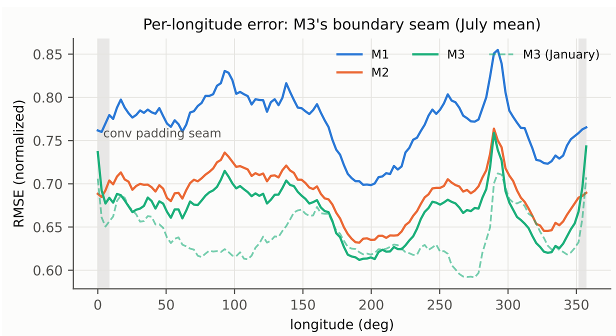

Figure 4: Month-mean RMSE by longitude. Only the padded model (M3) exhibits the boundary seam (shaded); the January overlay replicates it. Sources: `screen_p4_seam`, `a4_january`.

5.3 Finding 2: what the nonlocality is—and is not

Causally long-range. Occlusion gives a median saturation radius of 4 cells ($\sim$1200 km at T42; >16 cells for 4/20 targets). M2's context is information, not smoothing: clamping its neighborhood to nine copies of the center column makes it *worse than* M1 (gap recovery -0.70; consistent across 24 timesteps).

Not propagation-shaped. Our pre-registered flagship—that influence maps $I(x,y) = \sum_c |\partial E/\partial x_c(x,y)|$ (with E the predicted flux energy at a hotspot target) would be anisotropic along ray-traced propagation directions from our unit-tested tracer—was killed: median ring axis ratio 1.07 (threshold 1.3), wind alignment 9/20 (chance $\approx$6.7/20). At T42 the physically expected horizontal group displacements are largely sub-cell (Sec. 6), but the widely voiced "learned lateral propagation" narrative finds no support at the scale where these models operate.

Not simple. The M1$\to$M2 gap is closed by neither a linear model on the stencil (-47% of the gap—worse than M1) nor gradient features ($+9.6\%$; 3 seeds); per-neighbor resample-ablation shows no wind-conditioned directional structure above threshold (octant spread/mean $0.165 < 0.20$; an upstream-biased hint at best); and the context signal at the last hidden layer is high-rank (top-5 PCs $= 23\%$ of clamp-delta variance; projecting them out moves predictions 2.8% toward M1). The neighborhood is consumed as a full multivariate pattern.

Regime correlation (descriptive) and gates. The M2$\to$M3 gain concentrates in high-shear regions in both months (top-decile shear columns carry 2.9$\times$/1.8$\times$ their share) and avoids steep orography (0.27/-0.73), but the pre-registered confirmation threshold ($\geq$ 2.0) failed in January; we report the pattern descriptively. The attention gates are causally live (flattening the finest gate costs $+4.3\%$ RMSE, decaying monotonically with scale: $+1.4/+0.3/+0.1\%$) yet uncorrelated with instantaneous flux ($r = -0.015$ vs. nulls up to 0.07): modulation, not source selection. Mid-encoder features do carry a regime code that transfers across disjoint regions (orographic-vs-nonorographic AUC 0.72; random-init control 0.40; entire shuffled-label range below 0.54).

Table 4: Physics-graft battery. "Suppr." is flux suppression above the grafted reversal (Eq. 4); ρ is Spearman of response vs. amplitude; "align." is circular alignment of drag rotation with wind rotation. OOD exclusions per arm are reported in the repository (e.g., rotation: 145–159 of 280–320 rotations excluded).

Arm	Median response	IQR	N
Positive control (qbo1d emulator; ground truth known)			
Amplitude family (uncond.)	$\rho = 0.94$	[0.77, 0.94]	n=100
Reflection (effect $\geq$ 10%)	$r = 0.75$	[0.10, 0.83]	n=56
Column models (ERA5)			
M1 reflection, Jul (uvtheta)	suppr. -0.008	[-0.25, 0.26]	n=289
M1 reflection, Aug (uvthetaw)	suppr. 0.043	[-0.07, 0.30]	n=284
M1 amplitude (uvtheta / uvthetaw)	$\rho = 0.49$ / 0.49	–	n=162/136
M1 wind rotation (uvtheta / uvthetaw)	align. 0.22 / 0.23	–	n=135/121
Hotspot-center patch grafts (January)			
M3 patch (max dose)	suppr. 0.125	[0.05, 0.16]	n=28
dose direction	16/28 monotone, 12/28 anti		med. β=0.40
M1 paired (max dose)	suppr. 0.092		22/28 monotone

5.4 Finding 3: the physics-trust audit

Table 4 and Fig. 5 summarize the graft battery against the positive control.

Findings: (i) *typical columns show no filtering response* (median suppression -0.008, n=289, July uvtheta; $+0.043$, n=284, August uvthetaw—replicated across month and the released checkpoint/input-variant axis, which is confounded n=2); (ii) *amplitude response is non-monotone* (median $\rho = 0.49$ in both variants; testbed 0.94); (iii) *drag ignores wind rotation* (alignment 0.22/0.23; mean angular error 74–77°); (iv) *hotspot centers respond weakly and partially*: 10$\times$10-column patch reflections (January; 28 admissible patches) give median suppression 0.125 (M3) vs. 0.092 (paired M1) at max applied dose (median β 0.40; 20/28 ladders have only two distinct doses; dose direction mixed—M1 22/28 positive-monotone, M3 16/28 with 12 anti-monotone). This is far

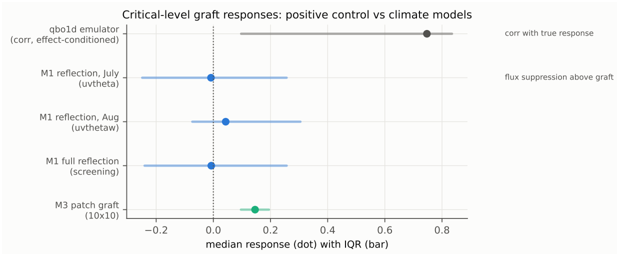

Figure 5: Critical-level graft responses. The validated positive control responds strongly (top row: correlation with the true response, conditioned on detectable effect); the column models do not (suppression medians at zero); hotspot-center patch grafts elicit only weak partial-dose responses. Sources: `gateweek_g1`, `d2_battery`, `screen_p_grafts`.

short of the near-complete flux removal linear theory prescribes at a full critical level, though per-patch quantitative expectations are not derived here (Sec. 6).

Internal evidence matches. No hidden layer carries linearly readable N^2 or Richardson number beyond raw inputs (negative selectivity at every depth against both controls); flux sign is barely decodable anywhere; the linear flux readout collapses after layer 1 and crystallizes in the last hidden layer (88% of full-model skill at the sixth hidden layer vs. 63% at the fifth)—consonant with the anchor's finding that last-two-layer fine-tuning suffices for transfer. Sparse autoencoders fail their pre-registered quantitative gates at both audited sites (codes remain dense, $L_0 \geq 106$ at all sparsity penalties tried; feature-map spatial coherence does not beat a random-direction null). Finally, the three architectures share a failure floor: top-1% errors co-occur at 38–53× chance (rank correlations 0.78–0.88), consistent with a common data-limited ceiling (ERA5's own unresolved GW fraction).

5.5 Hellinger is gameable where ladders are not

Two constructive demonstrations: variance-matched $\mathrm{M2}$ is physically absurd (variance ratio 11.9) yet achieves *better* pooled Hellinger (0.048) than $\mathrm{M3}$ (0.075); rolled $\mathrm{M3}$ loses its variance calibration entirely while pooled Hellinger is unchanged ($0.0745 \rightarrow 0.0744$). Pooled histograms reward marginal shape, not calibration; ladders (Eq. 2) discriminate in both directions. We recommend ladders alongside—or instead of—pooled Hellinger for distributional evaluation of flux surrogates.

6 Limitations

Scale. T42 coarse-grained fluxes; expected horizontal group displacements are largely sub-cell, so the propagation null may be resolution-conditional; conversely the position prior may be weaker for domains without fixed layouts. **Offline only**; no online/coupled claims (Pahlavan et al., 2024). **One checkpoint per configuration**; the checkpoint/input-variant robustness axis is confounded (n=2); no seed ensembles exist. **Roll attribution is an upper bound** (position-code removal and off-manifold degradation are inseparable with one checkpoint); four independent observations tie the collapse to position (symmetric distance curve; localization with exact identity control; hotspot concentration; $\mathrm{M1}/\mathrm{M2}$ controls; unchanged Hellinger). **Graft validity**: grafts break physical balance by design; admissibility and partial dosing bound but do not eliminate off-manifold effects; the positive control is 1D and column-local, covers the column-model arms (amplitude unconditionally; reflection under effect-detectability conditioning impossible in ERA5), and does *not* cover the U-Net patch arm—whose weak-response result is therefore bounded/descriptive; per-patch linear-theory expectations are future work with the released ray tracer. **Dose caps**: median applied β 0.40; 20/28 two-point ladders. **Labels**: ERA5's unresolved GW fraction bounds achievable physics (cf. the shared failure floor). **Epoch selection**: released epochs were presumably selected on 2015 validation loss; both audit months lie in 2015.

7 Conclusion

Audited mechanistically, the celebrated nonlocality of a released neural GW parameterization hierarchy resolves into a genuine but isotropic, regime-reading use of long-range context; a variance-calibration advantage carried substantially by an implicit, padding-derived geographic prior; and an absence of the causal wave physics the skill is often taken to imply—established with a positive-controlled intervention battery inside a pre-registered funnel in which most of our own hypotheses, including the flagship, died. None of this diminishes in-distribution utility; it sharpens what the skill *is*, how to measure it (ladders, not pooled Hellinger), which artifacts to fix (periodic padding; normalization metadata), and how much trust to extend where climate questions live: out of distribution.

Impact Statement

This work audits, rather than deploys, ML climate surrogates. Its findings support more trustworthy use of such models by identifying artifacts and evaluation blind spots, and its released tooling (conversion utilities, audit protocols) lowers the cost of similar audits. We see no direct negative societal impact; misreading our bounded negatives as a blanket claim against ML parameterization would be a misuse we explicitly disclaim (Secs. 1, 6).

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A The Hypothesis Funnel

Thirty hypotheses; verdicts and one-line evidence below. Full falsifiable statements, pre-registered kill criteria (with post-critic threshold amendments), and stamped per-verdict numbers are in the released `HYPOTHESIS_TABLE.md` and `results/`.

ID	Verdict	Evidence (one line)
N1	pass	finest-gate spatial structure 35× / temporal sensitivity 20× a random-init control
I2	pass	clamped neighborhood is worse than M1 (gap recovery -0.70); context is information
N3	pass	flattening finest gate: $+4.3\%$ RMSE, decaying $+1.4/+0.3/+0.1\%$ with scale
P4	pass	M3-only boundary seam $+5.1\%$ (Jul) / $+5.05\%$ (Jan); moves with padding under roll
A5	pass	calibration ladder splits hierarchy; all CIs disjoint (Table 2)
F3	pass	tail ordering matches Hellinger ordering (shared evidence with A5)
I4	pass	occlusion saturation median 4 cells; > 16 for 4/20 targets
R3	pass	regime code transfers across regions (AUC 0.72 vs. 0.40 rand-init)
R6	pass	flux lens collapses after layer 1, crystallizes at last hidden layer (88% of skill)
A1	kill	linear stencil closes -47% of the M1→M2 gap
A2	kill	shear concentration 1.38 $<$ 1.5 (July); spawned A2b
F2	kill	hotspot-type improvement ratio 1.06 (uniform)
N2	kill	finest gate vs. instantaneous flux $r = -0.015$; below climatology null
I1	kill	neighbor-ablation octant spread/mean 0.165 $<$ 0.20
A4	kill	clamp-delta top-5 PCs 23%; projection moves 2.8% toward M1
N5	kill	*flagship*: influence isotropy 1.07 $<$ 1.3; wind alignment 9/20
R1	kill	N^2/Ri probe selectivity negative at every depth
R2	kill	encoder reads z_s at R^2 0.06 vs. raw-input 0.45 (selectivity -0.39)
R5	kill	no sign-vs-magnitude decodability separation (sign selectivity $\leq$0.02)
P1	kill	suppression above grafted reversal -0.007 (median, n=277)
P2	kill	drag-rotation alignment 0.22; mean angular error 77°
P3	kill	amplitude response median $\rho = 0.49$; only 26% monotone
I3	kill	gradient features close 9.6% of the gap (3 seeds: 0.085–0.106)
R4	kill	SAE codes dense ($L_0 \geq 106$ at all λ); spatial coherence below random-direction null
F1	measured	top-1% errors co-occur at 38–53× chance; rank corr. 0.78–0.88
A3	deferred	reduced-scale retraining exceeded screening budget; causal half covered by N3
N4	deferred	conditional on N1 + an input-displacement null within budget
L1	deferred	transfer-learning checkpoints not verifiable within scope
L2	deferred	as L1
A2b	killed at confirmation	January shear concentration 1.77 $<$ 2.0 (pattern reported descriptively)

B Reproducibility

All experiments are config-driven and seeded; every results file embeds its config hash, seed, and library versions. Released: the hypothesis table with pre-registered criteria; the append-only research log (including failures, logged deviations, and three adversarial critic reports); the WxC normalization conversion utilities and their validation tests; physics baselines with 14 analytic unit tests (39 passing tests overall); `make reproduce-figures`. Only released artifacts (checkpoints, code, public data) are used. Compute: a single CPU laptop (i7-1065G7); the entire audit is reproducible without a GPU.